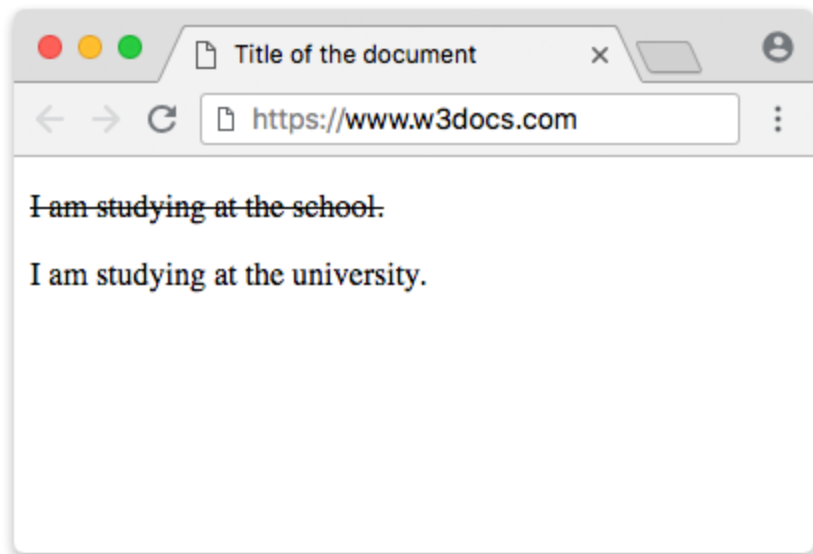




UNIT 3

- A **list** is a record of short pieces of related information used to display the data or any information in web pages in the ordered or unordered form.
- Lists are good from a structural point of view as they help create a well-structured, more accessible, easy-to-maintain document.
- For example, to purchase the items, we need to prepare a list that can either be ordered or unordered list which helps us to organize the data and easy to find the item.
- There are two different types of HTML lists:

1. Ordered List or Numbered List (ol)
2. Unordered List or Bulleted List (ul)

1. Ordered List or Numbered List (ol):

An ordered list is marked with the numbers by default. You can create an ordered list using the `<ol></ol>` tag and, define the list items using `<li></li>`.

We can create 4 types of ordered lists IN HTML –

- **type="1"**– This creates a numbered list starting from 1.
- **type="A"**– This creates a list numbered with uppercase letters starting from A.
- **type="a"**– This creates a list numbered with lowercase letters starting from a.
- **type="I"**– This creates a list numbered with uppercase roman numbers starting from I.
- **type="i"**– This creates a list numbered with lowercase roman numbers starting from i.

Syntax

```
<ol type="1">
  <li>Item in list...</li>
  <li>Item in list...</li>
  <li>Item in list...</li>
</ol>
```

Example 1:

```
<html>
<body>
<ol >
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>

</body>
</html>
```

Output:

1. Coffee
2. Tea
3. Milk

Example 2:

```
<html>
<body>
<ol type="A">
    <li>computer</li>
    <li>mouse</li>
    <li>keyboard</li>
</ol>
</body>
</html>
```

Output:

- A. computer
- B. mouse
- C. keyboard

2. UNORDERED LIST OR BULLETED LIST (UL):

- **HTML Unordered List** or Bulleted List displays elements in bulleted format . We can use unordered list where we do not need to display items in any particular order. The HTML ul tag is used for the unordered list.
- There can be 3 types of bulleted list:
 - disc
 - circle
 - square

Syntax:

 list of items

Example 1:

```
<html>
<body>
  <ul>
    <li>HTML</li>
    <li>Java</li>
    <li>JavaScript</li>
    <li>SQL</li>
  </ul>
</body>
</html>
```

Output:

- HTML
- Java
- JavaScript
- SQL

Example 2:

```
<html>
<body>
<ul style="list-style-type:square;">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
</body>
</html>
```

Output:

- Coffee
- Tea
- Milk

3. Nested list:

A **nested list** is simply a list inside another list. The proper way to create a nested list is to realize that the nested list is a child of a list item, not a list. A nested list can be ordered or unordered.

A. Nested unordered list

To create an unordered list in HTML, use the `ul` (unordered list) tag with the `li` (list item) tag in a hierarchical structure. In the example below, the list represents vegetable categories with a nested list showing the vegetable names of a particular category.

Example

```
<html>
<body>
<ul>
  <li>Fruit
    <ul>
      <li>Bananas</li>
      <li>Apples
        <ul>
          <li>Green</li>
          <li>Red</li>
        </ul>
      </li>
    </ul>
  </li>
</ul>
```

```
</li>
<li>Pears</li>
</ul>
</li>
<li>Vegetables</li>
<li>Meat</li>
</ul>
</body>
</html>
```

OUTPUT

- Fruit
 - Bananas
 - Apples
 - Green
 - Red
 - Pears
- Vegetables
- Meat

B. Nested ordered list

To create an ordered list in HTML, use the ol (ordered list) tag with the li (list item) tag in a hierarchical structure. In the example below, an ordered list represents the course curriculum with a nested list showing the subtopics of each course.

Example

```
<html>
<body>
  <h2>Nested Ordered List</h2>
  <ol>
    <li>Coffee</li>
    <li> Tea
      <ol>
        <li>Black tea</li>
        <li>Green tea</li>
      </ol>
    </li>
    <li>Milk</li>
  </ol>
</body>
</html>
```

OUTPUT:

Nested Ordered List

1. Coffee
2. Tea
 1. Black tea
 2. Green tea
3. Milk

Marquee HTML

The **Marquee HTML** tag is a non-standard HTML tag which was used to create a scrolling text or an image. It was used to make the text or image scroll horizontally across or vertically down the web page.

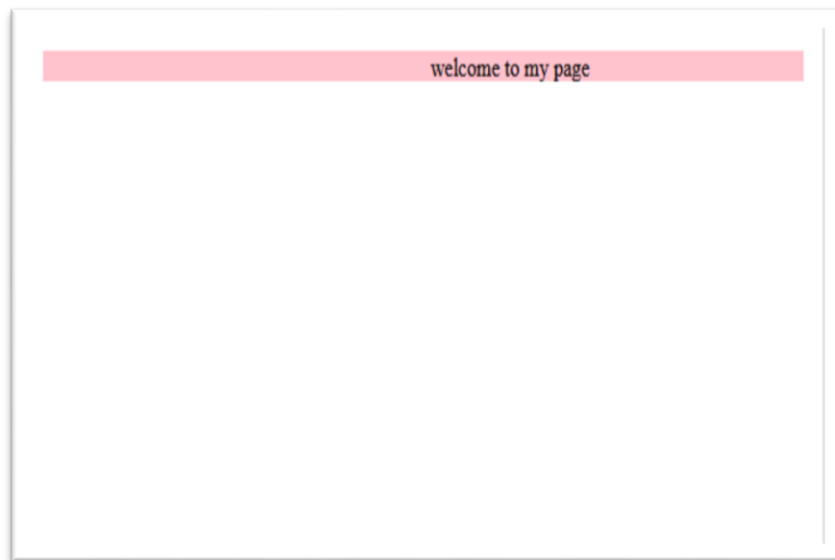
Syntax

```
<marquee> welcome </marquee>
```

Example 1:

```
<html>
<body>
<marquee width="100%" behavior="scroll" bgcolor="pink"> welcome to my page
</marquee>
</body>
</html>
```

OUTPUT:

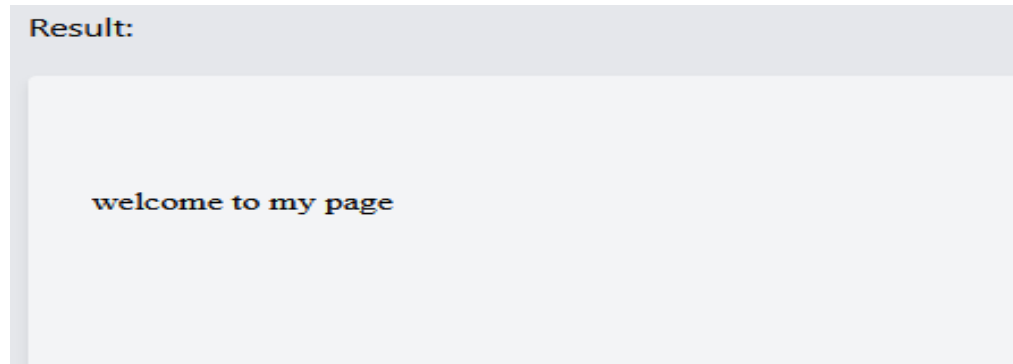


Example 2:

```
<html>
<head>
<title> marquee </title>
```

```
</head>
<body>
  <marquee direction="down">welcome to my page</marquee>
</body>
</html>
```

OUTPUT:



HR TAG :

- The `<hr>` tag in HTML is used to give a thematic break between paragraphs on a webpage.
- The `<hr>` tag is used when we want to change the topic or break the flow of the content.

Syntax

```
<hr>
```

EXAMPLE:

```
<html>

<head>

  <title>Title of the document</title>

</head>
```

<body>

<h1> What is the Internet</h1>

<p> Internet is a global network that connects billions of computers across the world with each other and to the World Wide Web. </p>

<hr>

<h1> What is HTML</h1>

<p> HTML stands for Hypertext Markup Language. It is the most basic language, and simple to learn and modify. It is a combination of both hypertext and markup language. </p>

</body>

</html>

OUTPUT:

What is the Internet

Internet is a global network that connects billions of computers across the world with each other and to the World Wide Web.

What is HTML

HTML stands for Hypertext Markup Language. It is the most basic language, and simple to learn and modify. It is a combination of both hypertext and markup language.

BR TAG:

- The
 tag inserts a single line break.
- The
 tag is useful for writing addresses or poems.

- The
 tag is an empty tag which means that it has no end tag.

Syntax

Text
 Text

Example:

```
<html>
```

```
<head>
```

```
<title> BR tag
```

```
</title>
```

```
</head>
```

```
<body>
```

```
<p>HTML stands for Hypertext Markup Language.<br> It is the most basic language, and  
simple to learn and modify.<br> It is a combination of both hypertext and markup language.  
</p>
```

```
</body>
```

```
</html>
```

Output:

HTML stands for Hypertext Markup Language.
It is the most basic language, and simple to learn and modify.
It is a combination of both hypertext and markup language.

IMAGE:

- The tag is used to embed an image in an HTML page.
- Images are not technically inserted into a web page; images are linked to web pages. The tag creates a holding space for the referenced image.

- That requires two attributes to be useful: src and alt.
- The **src** stands for ‘source’. The src attribute contains a URL pointing to the image you want to embed in the page.
- The **alt** stands for ‘alternate text,’ which is included by developers to entail what the image is about.

Images Syntax

```

```

EXAMPLE 1:

```
<html>

<head>

    <title>Image tag</title>

</head>

<body>

<h2><center>DEER</center></h2>



</body>

</html>
```

OUTPUT:

DEER



Creating Hyperlinks:

- HTML links are hyperlinks.
- If we want to add or insert the link in Html document for linking to another Html page with the current working page .
- You can click on a link and jump to another document.
- When you move the mouse over a link, the mouse arrow will turn into a little hand.

Syntax:

```
<a href="url">Link text</a>
```

Explanation:

<a>: This is the anchor tag that creates a hyperlink. Anything between this tag becomes part of the link. The user can click that part to reach the linked document.

- **href:** Specifies the destination address of the link used.
- **Text link:** The visible part of the link.

Example:

```
<html>

<head>

<title> web page linking</title>

</head>

<body>

<h1>HTML Links program</h1>

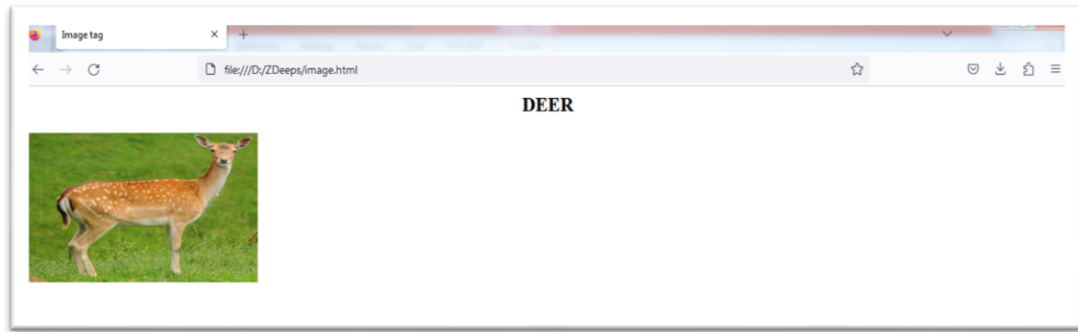
<p><a href=file:///D:/ZDeeps/image.html> image program </a></p>

</body>

</html>
```

Output:





UNIT 4

TABLE:

- A table is a structured set of data made up of rows and columns (**tabular data**).
- A table allows you to quickly and easily look up values that indicate some kind of connection between different types of data
- For example a person and their age, or a day of the week, or the timetable for a local swimming pool.

Person	Age
Chris	38
Dennis	45
Sarah	29
Karen	47

Table element in HTML

Table tag in HTML, is used to display data in tabular format (i.e. in row-column format). These table tags manages the layout of the page (i.e. Header, Body Content etc.).To create a table in HTML, you will need to define the table with the <table> tag. The <table> tag is the container of the table that specifies where the table will begin and where it ends.

Common HTML Table tags

- `<tr>` - represents rows
- `<td>` - used to create data cells
- `<th>` - used to add table headings
- `<caption>` - used to insert captions

Creating Tables in HTML

HTML table allows you to arrange data into rows and columns. They are commonly used to display tabular data like product listings, customer's details, financial reports, and so on. You can create a table using the `<table>` element. Inside the `<table>` element, you can use the `<tr>` elements to create rows, and to create columns inside a row you can use the `<td>` elements. You can also define a cell as a header for a group of table cells using the `<th>` element.

HTML Table Syntax:

```
<table>
  <tr>
    <td>Cell 1</td>
    <td>Cell 2</td>
    <td>Cell 3</td>
  </tr>
  <tr>
    <td>Cell 4</td>
    <td>Cell 5</td>
    <td>Cell 6</td>
  </tr>
</table>
```

Cell 1	Cell 2	Cell 3
Cell 4	Cell 5	Cell 6

Example 1:

```
<html>

<body>

<table border="1" style="width:40%" bordercolor = "green" bgcolor = "yellow" >

<h5> student details </h5>

  <tr>

    <th>No.</th>

    <th>Name</th>

    <th>Age</th>

  </tr>

  <tr>

    <td>1</td>

    <td>Peter Parker</td>

    <td>16</td>

  </tr>

  <tr>

    <td>2</td>

    <td>Clark Kent</td>

    <td>34</td>

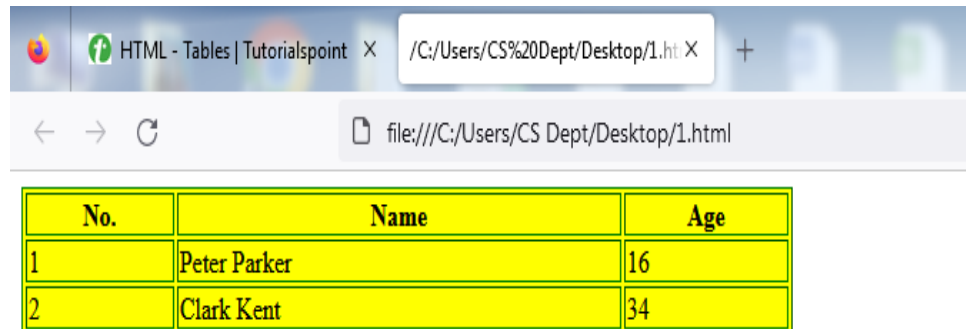
  </tr>

</table>

</body>
```

</html>

OUTPUT:



No.	Name	Age
1	Peter Parker	16
2	Clark Kent	34

CAPTION

The <caption> tag is used to define the header of the table. The tag itself must be inside the <table> element immediately after the opening (<table>) tag, and it must be the first child of its parent <table> element. It is possible to define only one caption per table.

When the <table> element containing <caption> is the only descendant of the <figure> element, you must use the <figcaption> element instead of <caption>.

By default, a table caption is center-aligned above a table. But it is possible to use the text-align and caption-side properties to align and place the caption.

Syntax

```
<caption> NAME LIST </caption>
```

EXAMPLE:

```
<html>

<body>

<table border="1" style="width:40%">

<caption >Employee Details</caption>

<tr>

<th>Firstname</th>

<th>Lastname</th>

<th>EmpId</th>

<th>Age</th>

<th>Designation</th>

</tr>

<tr>

<td>John</td>

<td>Roy</td>

<td>9111</td>

<td>28</td>

<td>Sales Head</td>

</tr>

<tr>

<td>Dinesh</td>

<td>Shetty</td>

<td>9112</td>

<td>43</td>
```

```
<td>Developer</td>
```

```
</tr>
```

```
<tr>
```

```
<tr>
```

```
<td>Niks</td>
```

```
<td>Raw</td>
```

```
<td>9114</td>
```

```
<td>45</td>
```

```
<td>Tester</td>
```

```
</tr>
```

```
</table>
```

```
</html>
```

OUTPUT:

Firstname	Lastname	EmpId	Age	Designation
John	Roy	9111	28	Sales Head
Dinesh	Shetty	9112	43	Developer
Niks	Raw	9114	45	Tester

Table & Cell Align Attribute

Table Align Attribute

The HTML **table align attribute** is used to control the horizontal alignment of a table on a web page. It specifies where the table should be positioned relative to the surrounding text or other content on the page.

The align attribute can be set to one of three values: left, center, or right. When set to **left** or **right**, the table will be aligned to the respective edge of the page, with any remaining space on the opposite side. When set to center, the table will be horizontally centered on the page.

The purpose of using the HTML **table align attribute** is to improve the visual appearance of a web page by aligning the table with the surrounding content in a way that is aesthetically pleasing and easy to read. This is especially important for pages that contain a lot of text and data, as it can be difficult for readers to navigate the content without clear visual cues.

For example, a table that is aligned to the left of a web page may be more visually appealing and easier to read when surrounded by text that is also **aligned to the left**. Conversely, a table that is centered on a page may be more effective in drawing attention to its content and separating it from surrounding text.

Syntax:

```
table align="left">
```

Attribute Values:

- **left:** It sets the left align to the table. It is a default value.
- **right:** It sets the right align to the table.
- **center:** It sets the center align to the table.

RIGHT ALIGNMENT

```
<html>
```

```
<head>
```

```

<title>
    HTML table align Attribute
</title>
</head>
<body>
    <h2>Table align Attribute</h2>
    <table border="1" align="right">
        <caption>STUDENT DETAILS</caption>
        <tr>
            <th>NAME</th>
            <th>AGE</th>
            <th>BRANCH</th>
        </tr>
        <tr>
            <td>UMA</td>
            <td>22</td>
            <td>CS</td>
        </tr>
        <tr>
            <td>RAM</td>
            <td>21</td>
            <td>ISM</td>
        </tr>
    </table>
</body>
</html>

```

OUTPUT

Table align Attribute

NAME	AGE	BRANCH
UMA	22	CS
RAM	21	ISM

LEFT ALIGNMENT

```
<html>
<head>
  <title>
    HTML table align Attribute
  </title>
</head>

<body>

  <h2>Table align Attribute</h2>

  <table border="1" align="left">
    <caption>STUDENT DETAILS</caption>

    <tr>
      <th>NAME</th>
      <th>AGE</th>
      <th>BRANCH</th>
    </tr>
    <tr>
      <td>UMA</td>
```

```

        <td>22</td>
        <td>CS</td>
    </tr>
    <tr>
        <td>RAM</td>
        <td>21</td>
        <td>ISM</td>
    </tr>
    <tr>
        <td>DEEPIKA</td>
        <td>21</td>
        <td>BCA</td>
    </tr>
</table>
</body>

</html>

```

OUTPUT

Table align Attribute

STUDENT DETAILS		
NAME	AGE	BRANCH
UMA	22	CS
RAM	21	ISM
DEEPIKA	21	BCA

CENTER ALIGNMENT

```
<html>
<head>
  <title>
    HTML table align Attribute
  </title>
</head>

<body>

  <h2>Table align Attribute</h2>

  <table border="1" align="center">
    <caption>STUDENT DETAILS</caption>

    <tr>
      <th>NAME</th>
      <th>AGE</th>
      <th>BRANCH</th>
    </tr>
    <tr>
      <td>UMA</td>
      <td>22</td>
      <td>CS</td>
    </tr>
    <tr>
      <td>RAM</td>
      <td>21</td>
      <td>ISM</td>
    </tr>
    <tr>
```

```
<td>DEEPIKA</td>
<td>21</td>
<td>BCA</td>
</tr>
</table>
</body>

</html>
```

OUTPUT

Table align Attribute

NAME	AGE	BRANCH
UMA	22	CS
RAM	21	ISM
DEEPIKA	21	BCA

Cell Align Attribute :

Horizontal Alignment

The text-align property sets the horizontal alignment (like left, right, or center) of the content in <th> or <td>.By default, the content of <th> elements are center-aligned and the content of <td>

elements are left-aligned. To center-align the content of `<td>` elements as well, use `text-align: center`

EXAMPLE:

```
<html>
<body>
<table border="1" width="500" >
  <tr style="text-align:center;">
    <th>name</th>
    <th>age</th>
  <tr style="text-align:right;">
    <td>uma</td>
    <td>55</td>
  </tr>
  <tr style="text-align:left;">
    <td>siva</td>
    <td>44</td>
  </tr>
  <tr style="text-align:center;">
    <td>kumar</td>
    <td>44</td>
  </tr>
</table>
</body>
</html>
```

OUTPUT:

name	age
uma	55
siva	44
kumar	44

Vertical Alignment

The vertical-align property sets the vertical alignment (like top, bottom, or middle) of the content in <th> or <td>. By default, the vertical alignment of the content in a table is middle (for both <th> and <td> elements). The following example sets the vertical text alignment to bottom for <td> elements:

PROGRAM

```
<html>

<body>

<table border="1" width="100%" height="100%">

  <tr>

    <th valign="top">Header 1</th>

    <th valign="middle">Header 2</th>

    <th valign="bottom">Header 3</th>

  </tr>

  <tr>

    <td valign="top">Row 1, Column 1</td>
```

```

        <td valign="middle">Row 1, Column 2</td>

        <td valign="bottom">Row 1, Column 3</td>

    </tr>

    <tr>

        <td valign="top">Row 2, Column 1</td>

        <td valign="middle">Row 2, Column 2</td>

        <td valign="bottom">Row 2, Column 3</td>

    </tr>

</table>

</body>

</html>

```

OUTPUT:

Header 1	Header 2	Header 3
Row 1, Column 1	Row 1, Column 2	Row 1, Column 3
Row 2, Column 1	Row 2, Column 2	Row 2, Column 3

colspan

Colspan in HTML is an attribute that defines the number of columns a single cell can span in a table.

It allows the single table cell to span the width of more than one cell or column.

It provides the same functionality as “merge cell” in a spreadsheet program like Excel.

It can be used with <td> and <th> element while creating an HTML Table.

Attribute Values:

<td>: The colspan attribute when used with <td> tag determines the number of standard cells it should span.

Syntax:

```
<td colspan = "value">table content...</td>
```

Example 1:

```
<html>

<body>

<h2>student details</h2>

<table border="1" style="width:40%">

  <tr>

    <th colspan="2">Name</th>

    <th>Age</th>

  </tr>

  <tr>
```

```

<td>siva</td>

<td>raja</td>

<td>43</td>

</tr>

<tr>

<td>ram</td>

<td>arun</td>

<td>57</td>

</tr>

</table>

</body>

</html>

```

Output:

student details		
Name		Age
siva	raja	43
ram	arun	57

rowspan:

The rowspan attribute in HTML specifies the number of rows a cell should span.

That is if a row spans two rows, it means it will take up the space of two rows in that table. It allows the single table cell to span the height of more than one cell or row.

It provides the same functionality as “merge cell” in the spreadsheet program like Excel.

Attribute Values:

- **<td>:** The rowspan attribute when used with <td> tag determines the number of standard cells it should span.

Syntax:

```
<td rowspan = "value">table content...</td>
```

EXAMPLE:

```
<html>
```

```
<body>
```

```
<h2>Student phone number</h2>
```

```
<table border="1" style="width:40%">
```

```
<tr>
```

```
<th>Name</th>
```

```
<td>Deepika</td>
```

```
</tr>
```

```
<tr>
```

```
<th rowspan="2">Phone</th>
```

```
<td>9632587410</td>

</tr>

<tr>

<td>9874663210</td>

</tr>

</table>

</body>

</html>
```

Output:

Student phone number

Name	Deepika
Phone No	9632587410
	9874663210

CELLPADDING

cellpadding is use to adjust the white space in your table cells. The cellpadding represents the distance between cell borders and the content within a cell.

Cell padding is the space between cell borders of the table and the content within a cell of the table.

Syntax:

```
<table cellpadding="pixels">
```

Example

```
<html>
  <head>
    <title>HTML Table Cellpadding</title>
  </head>
  <body>
    <table border = "1" cellpadding = "50" cellspacing = "5">
      <tr>
        <th>Name</th>
        <th>Salary</th>
      </tr>
      <tr>
        <td>Ramesh Raman</td>
        <td>5000</td>
      </tr>
      <tr>
        <td>Shabbir Hussein</td>
        <td>7000</td>
      </tr>
    </table>
  </body>
</html>
```

Output:

Name	Salary
Ramesh Raman	5000
Shabbir Hussein	7000

CELLSPACING

Cellspacing is something different from cellpadding. In HTML, cellspacing is yet another attribute of the table tag. It regulates the distance between the single cells in a table. By using this feature, developers might simply change the space between the edges of several adjacent cells. It enhances the table's readability. The cell border appears to increase as the number of cellspacing increases.

Cellspacing is most commonly utilized in web design. Different programming languages utilize different syntax for cellspacing, like CSS utilizes the "***border-spacing***" property, whereas HTML utilizes the "***cellspacing***" property.

Syntax:

```
<table cellspacing="value" >.....</table>
```

EXAMPLE:

```
<html>
<head>
  <title>HTML Table Cellpadding</title>
</head>
```

```
<body>
  <table border = "1" cellpadding = "30" cellspacing = "5">
    <tr>
      <th>Name</th>
      <th>Salary</th>
      <th>phone no</th>
    </tr>
    <tr>
      <td>deepiks</td>
      <td>5000</td>
      <td>963258741</td>
    </tr>
    <tr>
      <td>Shabbir Hussein</td>
      <td>7000</td>
      <td>123654789</td>
    </tr>
    <tr>
      <td>uma</td>
      <td>7000</td>
      <td>523654789</td>
    </tr>
  </table>
</body>
</html>
```

OUTPUT: